

Reading Matching Results

Module 2 · Day 2, Session 3: Two Case Studies, One Clean and One Not

3ie

1 • The third way to build a comparison

Where we are

- Day 1 S3** randomisation · the comparison chosen by a lottery
- Day 2 S1** difference-in-differences · and an assumption that could not be tested
- Day 2 S2** regression discontinuity · the comparison the rule creates
- Today** matching · the comparison you build yourself

Four methods, four ways of answering **compared to what?**

- Randomisation **creates** a comparison group.
- DiD uses the same groups' **change over time** (-816 AED this morning).
- A cut-off **creates** one at the line (-791 AED).
- Matching **constructs** one, by finding lookalikes.

DiD said -816, RDD -791, both short of 1,000. Matching uses every business. Does it change the verdict?

[MENTI PLACEHOLDER] Would that be fair?

Two businesses, one enrolled and one did not. They are the same size, the same age, the same area, in the same kind of neighbourhood.

Is comparing them fair?

Yes · No · Only if nothing else matters

Go to **menti.com** and enter code **1234 5678**, or scan the code.



“Only if nothing else matters” is the best answer. Most of this session explains why.

Matching handles what you measured. Everything you did not measure is left in the comparison, and you will not be able to see it in the table.

Matching, in one sentence

**For each business that got the programme,
find one that did not but looks just like it.
Then compare the two.**

There is no lottery and no cut-off. You build the comparison group yourself, one matched pair at a time, using the characteristics you measured **before** the programme started.

Say it back to your neighbour.

Your turn: find the twins

Enrolled			
	Manager age	Size	Cost after
A	45	small	900
B	38	large	1,400
C	52	small	700

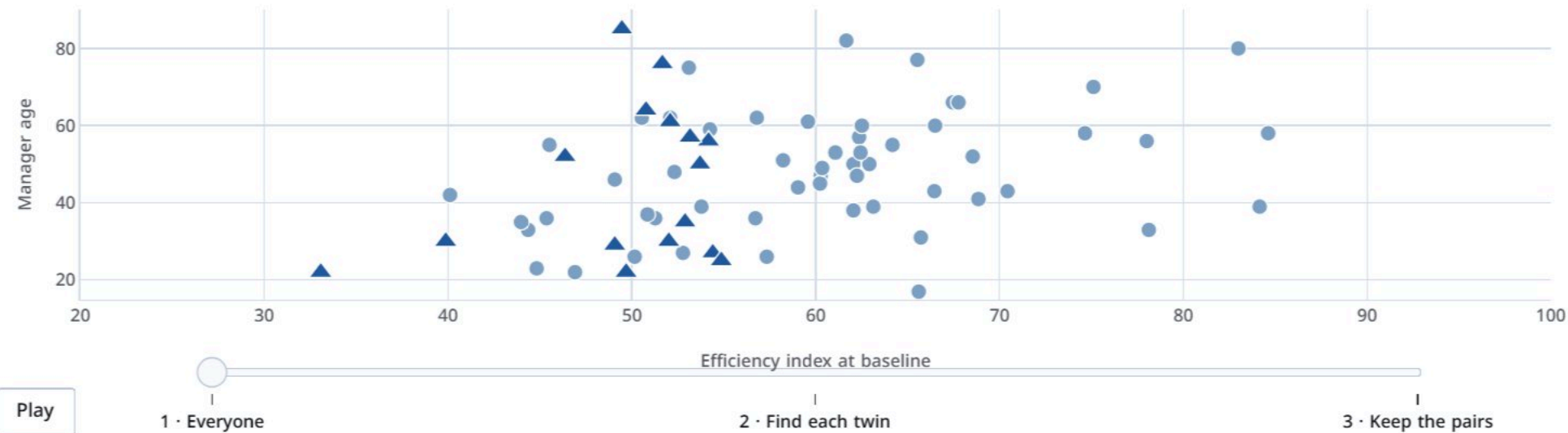
Not enrolled			
	Manager age	Size	Cost after
1	39	large	2,300
2	46	small	1,900
3	51	small	1,800
4	29	large	2,600

Pair each enrolled business with its closest twin, matching on **who they are**, not on their cost. Then take the average cost difference across the three pairs. Ninety seconds.

A with 2, B with 1, C with 3. Differences of -1,000, -900 and -1,100: an average of **-1,000 AED**. Business 4 is nobody's twin, so it is set aside.

What matching does, on real businesses

▲ enrolled ● not enrolled (71 real GreenWaste businesses, matched on two characteristics)



Drag the slider. Where do the triangles sit, and which circles are never anyone's twin?

2 · Implementing matching in R

Remember: what matching needs from the data

1. **An outcome**, measured the same way for everyone
2. **A group that got the programme and a pool that did not**, and a variable saying which is which
3. **Characteristics measured before the programme**, never after it
4. **The characteristics that drove who took part**, and that also affect the outcome
5. **Enough untreated units that look like the treated ones**, so every treated unit has a real twin

Which of the five would you check first in your own data?

Take home: ask AI *what data do I need for matching, and does my dataset have it?*, and have it question you about who took part before it answers.

Does GreenWaste have what we need?

```
R Code ↺ Start Over ▶ Run Code
1 biz %>%
2   group_by(enrolled) %>%
3   summarise(businesses = n(),
4             cost_before = round(mean(y0)),
5             cost_after  = round(mean(y1)),
6             efficiency  = round(mean(efficiency_index), 1),
7             staff       = round(mean(staff_size), 1))
```

```
# A tibble: 2 × 6
  enrolled businesses cost_before cost_after efficiency staff
  <int>      <int>      <dbl>      <dbl>      <dbl> <dbl>
1      0      6949      1837      2071       60    4.9
2      1      2964      1449       784      49.3    5.8
```

- ✓ **Outcome, group:** `y1` costs after; `enrolled` 1 or 0
- ✓ **Before the programme:** ten round-0 characteristics, incl. the efficiency index
- ✓ **A big pool:** 6,949 possible twins for 2,964 enrolled

The enrolled were already cheaper **before** the programme. The raw gap after, **-1,287 AED**, mixes the programme with who they already were.

Implementing matching: four steps

```

1 # Step 1. A score: how likely was each business
2 #       to enrol, given what we knew at baseline?
3 fit <- glm(enrolled ~ efficiency_index + age_manager +
4           educ_manager + female_manager + foreign_owned +
5           staff_size + advanced_filtration +
6           water_treatment_system + business_area +
7           recycling_center_distance,
8           data = biz, family = binomial)
9
10 # Step 2. Each enrolled business takes the
11 #         not-enrolled business with the closest score.
12 # Step 3. Check the pairs are alike (balance).
13 # Step 4. Compare their costs after the programme.
14 lm_robust(y1 ~ enrolled, data = matched,
15           clusters = neighborhood_idenfifier)

```

- **Step 1** squeezes ten characteristics into one number between 0 and 1, the **likelihood of enrolling**
- **Step 2** is your twins exercise, done 2,964 times
- **Step 3** is the only evidence the pairs are alike
- **Step 4** is the comparison you wanted all along

The **list in Step 1** is the analyst's choice. Section 3 tests it.

Steps 1 and 2, live

R Code

↺ Start Over

▶ Run Code

```

1 # Step 1. The likelihood-of-enrolling score
2 fit <- glm(f_all, data = biz, family = binomial)
3 biz$score <- fitted(fit)
4
5 # Step 2. Each enrolled business takes the
6 #         closest score among the rest
7 took <- filter(biz, enrolled == 1)
8 rest <- filter(biz, enrolled == 0)
9 twin <- sapply(took$score, function(s)
10               which.min(abs(s - rest$score)))
11 matched <- bind_rows(took, rest[twin, ])
12
13 c(enrolled    = nrow(took),
14   twins_used  = length(unique(twin)))

```

enrolled	twins_used
2964	1619

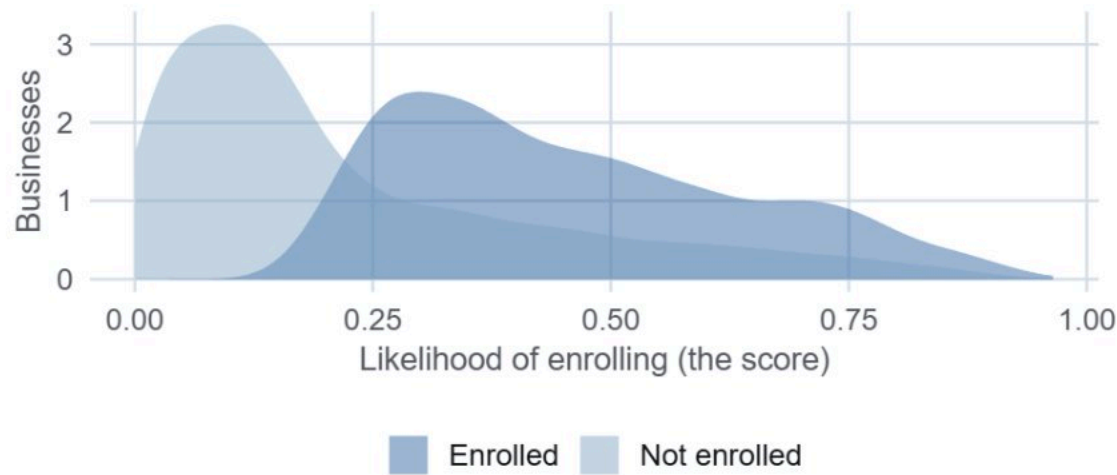
Guess first: 2,964 different twins, or far fewer? Then press **Run**.

1,619 twins for 2,964 enrolled businesses. Some not-enrolled businesses are the closest match for several enrolled ones, and are used more than once.

The other 5,330 not-enrolled businesses are set aside. Business 4, 5,330 times over.

Did every enrolled business find a real lookalike?

```
R Code Start Over Run Code
1 biz %>%
2   mutate(group = ifelse(enrolled == 1,
3                         "Enrolled", "Not enrolled")) %>%
4   ggplot(aes(score, fill = group)) +
5   geom_density(alpha = 0.45, colour = NA) +
6   scale_fill_manual(values = grp_cols) +
7   labs(x = "Likelihood of enrolling (the score)",
8        y = "Businesses") +
9   theme_m2()
```

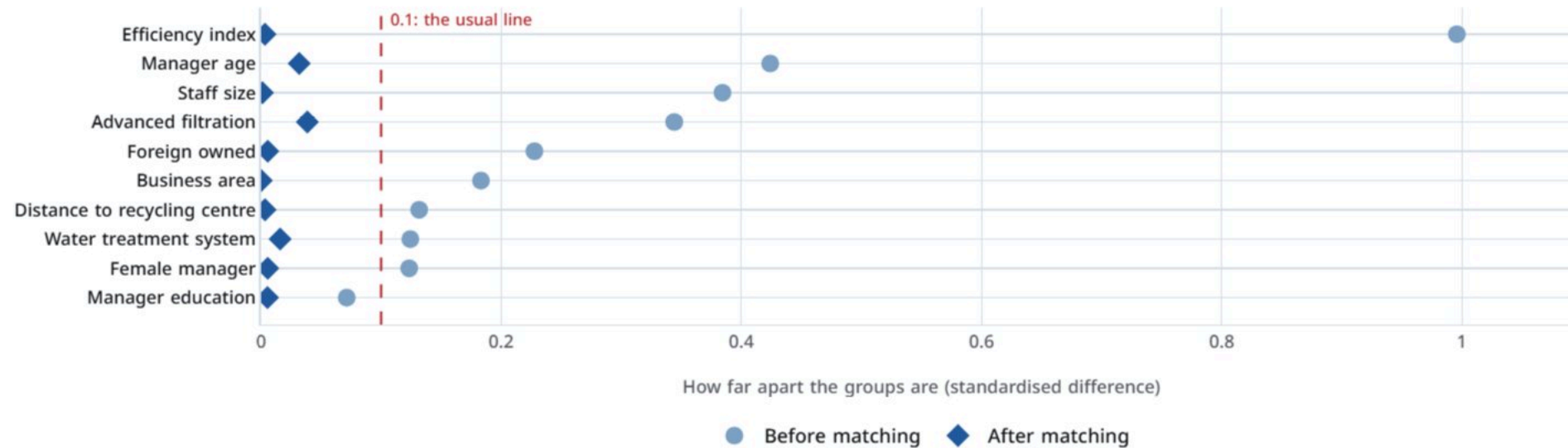


A twin can only come from where the curves **overlap**. Out past every not-enrolled business, there is no real lookalike, only the least bad one.

Here the ranges overlap almost completely. Only **1 of 2,964** enrolled businesses scores beyond the pool.

What matters is **which** units have no match. If they are a particular kind, the result no longer covers that kind. Case 2 shows this.

Did matching work? Balance



Each row is one characteristic. Hover to read the values. What happened to the efficiency index?

Step 4: running the comparison

R Code

Start Over

Run Code

```

1 out <- lm_robust(y1 ~ enrolled, data = matched,
2                 clusters = neighborhood_identifier)
3
4 res <- tidy(out)[, c("term", "estimate", "std.error",
5                     "conf.low", "conf.high")]
6 res[, -1] <- round(res[, -1])
7 res

```

	term	estimate	std.error	conf.low	conf.high
1 (Intercept)		1782	36	1710	1854
2 enrolled		-998	44	-1085	-911

Two rows will appear. Guess first: which one is the programme's effect? Then press **Run**.

Row 2, enrolled: -998 AED. The cost difference between each enrolled business and its twin, averaged.

Trap: the intercept, 1,782 AED, is the twins' average cost, not an effect.

Does it clear the rule?

-998

AED, the matched estimate (95% CI: -1,085 to -911)

Before matching **-1,287** The rule **1,000** Least it could be **911**

Matching took 289 AED off the raw gap: that part was **who the businesses already were**.

Does the whole interval clear 1,000 AED?

No. The estimate sits right on the bar, and the interval runs from 911 to 1,085 AED. A result this close to the line turns on the analyst's choices, which we test next.

Five-minute break

Back in five minutes. When we return: what matching cannot see, and how to check how much that matters.

3 · The one question, and how to check it

What matching can and cannot see

Matching can only account for what you measured.

Every characteristic in that balance chart was **collected**. Matching cannot balance on management quality, on how motivated the owner was, on whether the business was about to invest anyway, unless somebody wrote it down.

So the accurate claim for this method is: **we have made the two groups alike in every way we can observe.**

Check 1: what if we had never measured the index?

R Code

↻ Start Over

▶ Run Code

```

1 # Re-run Steps 1 and 2, pretending the
2 # efficiency index was never collected
3 biz$score2 <- fitted(glm(f_no_index, data = biz,
4                          family = binomial))
5 m2 <- find_twins(biz, "score2")
6
7 m2 %>%
8   group_by(enrolled) %>%
9   summarise(cost_after = round(mean(y1)),
10             index      = round(mean(efficiency_index), 1),
11             staff      = round(mean(staff_size), 1),
12             manager     = round(mean(age_manager), 1))

```

A tibble: 2 × 5

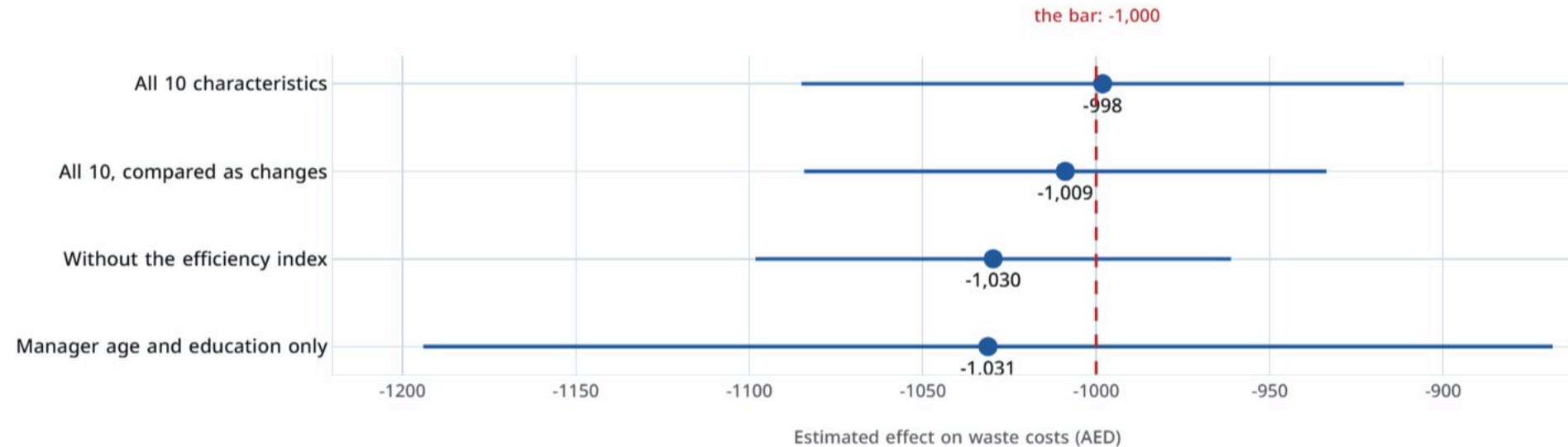
	enrolled	cost_after	index	staff	manager
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	1814	55.8	5.6	41.6
2	1	784	49.3	5.8	41.7

Press **Run**. The index is no longer in the recipe. Before you look: will the pairs still match on it?

The pairs are now **6.5 index points apart**, and the estimate moves to **-1,030 AED**, across the bar.

The balance check on the nine would still pass (worst 0.06). **Nothing in the output would warn you.**

Check 2: change the recipe, see what moves



Same businesses, same method, four choices of list. The estimates land on **both sides of the bar**, and every interval crosses it.

Would you sign off a national scale-up on any single one of these rows?

The question to put to any evaluator

For each business that enrolled, you found one that did not but looked the same.

What else might be different about those two businesses that you could not measure?

It cannot be answered from the table.

For GreenWaste, the plausible answers are things like:

How badly did this business want to cut costs? Businesses that enrol are often the ones already looking for a solution.

Was the owner about to invest anyway?
If so, we are crediting the programme with a decision already made.

Neither of these is in the data. Both would make the programme look better than it was, just as leaving out the index did.

4 · Case 2: when the worst roads have no lookalike

The cameras went where the roads were worst

Has a camera	Segments	85th pct speed	Collisions, baseline	Collisions, prior 3 yrs
No	805	66.4	9.52	10.9
Yes	395	78.7	12.32	13.1

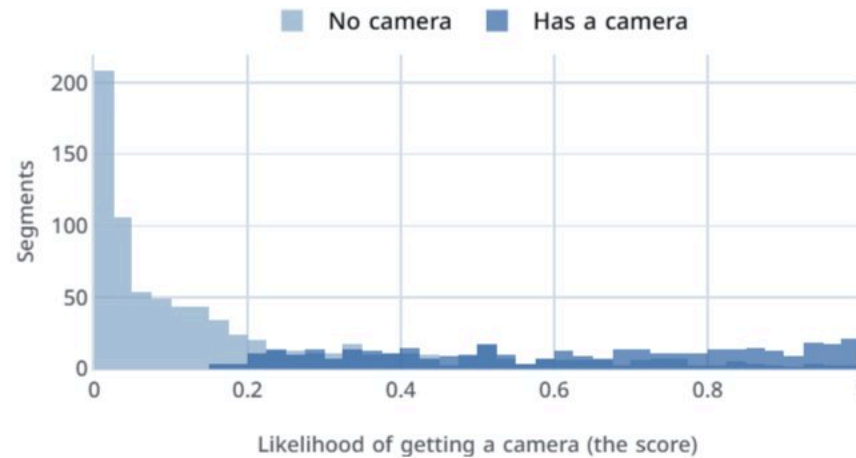
Illustrative case, with invented figures. The roads-authority data from Day 1; not a real programme.

You met these cameras on Day 1. The camera segments were **faster** and had **more collisions** before anything was installed.

In GreenWaste, businesses opted in. Here the authority chose the worst roads on purpose.

Same four steps, same checks. Rule: at least **2.0 fewer collisions** per segment per round. Which check do you expect to fail?

Check: is there a lookalike for every camera segment?



Camera segments pile up at the **high** end of the score, just where no-camera segments thin out.

13 camera segments (3%) score higher than any segment without a camera.

Few, but they are the **fastest** camera roads (96 km/h against 78 for the rest), the ones the cameras were aimed at. The software still paired them, with the least bad one.

And 395 camera segments share **171 twins**. The ten most-used twins stand in for 21% of all pairs (GreenWaste: 3%).

Check: mostly balanced, but not level

Characteristic	Before	After
85th percentile speed (km/h)	1.29	0.00
Lanes	0.01	-0.12
Segment length (km)	0.03	-0.04
School within 500 m	0.07	-0.07
Collisions, prior 3 years	0.41	-0.03
Poor lighting	0.12	0.17
Collisions at baseline (the outcome)	0.48	0.20

Standardised differences. Beyond about 0.1 is usually a problem.

Most characteristics in the model are within 0.1; 2 are not (poor lighting 0.17, lanes 0.12). The **outcome itself**, measured before the cameras went in, is furthest apart: **12.32** collisions on camera segments against **11.18** on their twins.

Matching balanced most of what the model **contained**. Collisions in the baseline year were not in it, so they were free to differ, and they sit furthest over 0.1.

Three numbers from the same pairs

2.10

collisions avoided per segment, comparing changes (95% CI: 1.32 to 2.88)

Comparing follow-up levels **0.96** Plain DiD, no matching **1.81** The rule **2.0**

Levels says 0.96 avoided and fails the rule; **changes** says 2.10 and clears it. Which do you believe?

Changes. The pairs started 1.15 collisions apart, and changes remove that. It lands near the plain DiD. **But** the least generous end, 1.32, is short of 2.0.

The two cases, side by side

	Case 1 · GreenWaste	Case 2 · Cameras
Every treated unit had a real lookalike?	yes, all but 1	no, 13 segments (3%) outside
Comparison units reused as twins	1,619 twins for 2,964 businesses	171 twins for 395 segments
Characteristics balanced after matching?	yes (worst 0.04)	partly: 2 of 6 over 0.1 (worst 0.17)
Outcome's own baseline balanced?	yes (gap 11 AED)	no (std. diff 0.20)
Estimate, follow-up levels	-998 AED	0.96 avoided
Estimate, changes	-1,009 AED	2.10 avoided
Least generous end vs the rule	911 against 1,000: short	1.32 against 2.0: short

Read down the table. Which rows separate the two cases, and would you see them in a typical results table?

5 · Group exercise: which would you trust?

Which case would you trust?

In your groups, ten minutes:

1. Which case gives you **more confidence**, and why?
2. What would you ask **Case 2's** evaluator that Case 1 does not raise?
3. Case 2's changes estimate clears its rule by more than Case 1's does (5% against 1%). Does that change your answer?

Our read: **Case 1** is the more trustworthy study, and it cannot show it clears the rule. **Case 2** gives the more attractive answer, and it carries the unquantified risk.

What to ask the evaluator of Case 2

- Ask** "What happened to the segments with no real lookalike, and does the answer change without them?"
- Ask** "Did the outcome's own baseline balance, and did you compare levels or changes?"

6 • Put AI to work

AI Snapshot: comparing two studies

Prompt, with the two-case table and both balance tables pasted in: “Summarise the key differences between these two studies and identify what a commissioner should flag.”

A response like this is possible, especially with a short or leading prompt:

Case 1: Matching was highly effective, with every business matched and excellent covariate balance. The programme reduced costs by around 1,000 AED, with an interval that straddles the 1,000 AED threshold.

Case 2: Matching was similarly strong, with every segment matched and good balance. The matched estimate shows cameras avoided 2.10 collisions per segment, comfortably exceeding the 2.0 threshold, so this programme should be scaled.

Which sentences would you sign your name to? Four minutes, in groups.

Debrief: one clean read, three false assurances

What it got right. Case 1 is reported accurately, including the awkward part: the interval straddles the rule.

1 • “Similarly strong” in Case 2. It was not. 13 segments had no real lookalike, and “every segment matched” hides that.

2 • “Comfortably exceeding.” Only the changes-based estimate clears 2.0; the levels estimate does not, and the interval dips to 1.32.

3 • “So this programme should be scaled.” The recommendation rests entirely on sentences 1 and 2, and it is stated most confidently of all.

The model did not invent a number. It had the details in front of it and still went with the headline. Check its summary against the table you gave it.

The same question, a better prompt

What people actually type

Summarise the key differences between these two studies and identify what a commissioner should flag.

The model may summarise the tables it was given and repeat their framing.

What to type instead

Two matching studies. For each: how many treated units had **no real lookalike**; did the **outcome's baseline** balance; do levels and changes agree? Our decision rules are below. **Ask me questions before you answer.**

Take home: run both after the session. What did each get wrong, and what did each explain well?

7 • Close

Your turn: ask AI first

Before we show our list of questions to ask someone presenting matching results, ask an AI tool:

“I am a commissioner about to hear an evaluator present a matching result. What are the key questions I should ask?”

Three minutes. Then compare with our list on the next slide: what did it miss, what did it add, and is anything incorrect?

Five questions to ask any evaluator presenting matching

- 1 What did you match on, and what was left out?
- 2 How many treated units had no real lookalike, and what happened to them?
- 3 Do the covariates balance, and does the outcome's baseline balance?
- 4 Does the answer hold if you change what you matched on, or compare changes instead of levels?
- 5 What else might differ that you could not measure?

Question 5 is the one that matters most, and it is the only one the evaluator cannot answer from their own output.

So a matching study rarely settles a decision on its own.

The week so far

- Day 1 S2** descriptive reading · -2.53 against a 2.0 rule, with no comparison group
- Day 1 S3** randomisation · -1,014, interval spans 1,000
- Day 2 S1** DiD · -816 against a 1,000 rule, and an assumption that could not be tested
- Day 2 S2** RDD · -791 against a 1,000 rule, and local to the line
- Today** matching · -998 against a 1,000 rule, and only as good as the list

Each method rests on one assumption:

DiD assumes parallel trends. **RDD** assumes nothing else jumps at the cut-off, and answers only near it. **Matching** assumes everything that matters was measured.

Which of the three would you trust least, and why?